

National Institute of Technology Calicut

Department of Computer Science and Engineering

Monsoon Semester 2026 – 2027

Course Plan : CS3001E Computer Networks

Credits: 4

Class: B. Tech. (V semester)

Slot: C2 and C2+

Lecture & Tutorial Hours

- **Monday** (C2: 4:00 PM to 4:50)
- **Tuesday** (C2: 3:00 PM to 3:50 PM)
- **Wednesday** (C2: 2:00 PM to 2:50 PM)
- **Friday** (C2+: 1:00 PM to 1:50 PM)

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Online Resources through : NITC Eduserver <https://moodle.nitc.ac.in/login/index.php>.

Enrollment Key : aiwf2q

Course Outcomes

- **CO1:** Analyse and evaluate datalink, network, transport, and application layer protocols
- **CO2:** Apply the theory of basic networking concepts in the performance analysis of the protocol stack
- **CO3:** Design, implement, and analyse solutions to problems in various layers of the protocol stack.

Brief Syllabus

- Computer Networks and Internet, The network edge, The network core, Network access, Delay and loss, Protocol layers and services, Application layer protocols, Web 2.0, Web 3.0, Socket Programming.
- Transport layer services, UDP, TCP, New transport layer Protocols, Congestion control, New versions of TCP, Network layer services, Routing, Software Defined Networking, Openflow, P4 programming, IPv4, Routing in Internet, Router, IPV6, Multicast routing.
- Link layer services, Error detection and correction, Multiple access protocols, ARP, Ethernet, Hubs, Bridges, Switches, MPLS, VLAN.
- Wireless links, Mobility, Visible light communication techniques, Multimedia networking, Streaming stored audio and video, Real-time protocols, Network management.

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Reference Books

1. J. F. Kurose and K. W. Ross, Computer Networking: A Top-Down Approach Featuring Internet, 6/e, Pearson Education, 2012.
2. L. L. Peterson and B. S. Davie, Computer Networks, A systems approach, 5/e, Morgan Kaufmann, 2011.
3. A. S. Tanenbaum and D. J. Wetherall, Computer Networks, 5/e, Pearson, 2013.

Evaluation Scheme with Tentative dates

- Midterm : 30 Marks (8, September 2026)
- Quiz Evaluation : 30 Marks (30, October 2026)
- End Semester : 40 Marks (13, November 2026)

Attendance

- Minimum 70% (Strictly Follow)

Grading Policy

- Grading will be relative
- Makeup examination for midterm will be given only in genuine cases where prior (atleast one day before the midterm exam) intimation is submitted to the course faculty for the consent through faculty advisor
- All issues regarding the valuation of the midterm exams and term projects must be resolved within two days after the marks are announced

Standard of Conduct

Each student is expected to adhere to high standards of ethical conduct, especially those related to cheating. Any academic dishonesty will result in zero marks in the corresponding exam or quiz and will be reported to the department council for record keeping and for permission to assign F grade in the course. CSE Department policy on academic integrity is available at: <http://minerva.nitc.ac.in/cse/sites/default/files/attachments/news/Academic-Integrity.pdf>

Tentative Week-wise Course Plan (13 Weeks)

- **Week 1:** Introduction to Computer Networks and the Internet, The Network Edge, The Network Core, Delay, Loss and Throughput
- **Week 2:** Protocol Layers and Services, Application Layer: Client-server, Peer-peer architecture
- **Week 3:** Web, HTTP/HTTPS, Cookies, Web Caching
- **Week 4:** DNS, E-mail Protocols (SMTP, POP3, IMAP)

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- **Week 5:** Multimedia Applications, Streaming Applications, Socket Programming using TCP and UDP
- **Week 6:** Transport Layer Services, Multiplexing and Demultiplexing, UDP, Reliable Data Transfer Principles.
- **Week 7:** TCP Fundamentals, Reliable Data Transfer, Flow Control, TCP Connection Management
- **Week 8:** TCP Congestion Control, TCP Performance, New Versions of TCP, Emerging Transport Protocols (e.g., QUIC)
- **Week 9:** Network Layer Services, Router Architecture, IPv4 Addressing, Subnetting, Forwarding and Routing
- **Week 10:** Routing Algorithms, Routing in the Internet, IPv6, Multicast Routing
- **Week 11:** Link Layer Services, Error Detection and Correction, Multiple Access Protocols, ARP, Ethernet.
- **Week 12:** Hubs, Bridges, Switches, VLAN, MPLS, Switched LANs
- **Week 13:** Wireless links, Mobility